Addendum to Release Notes for 3.0.10, 3.1.2, and 3.2.1 code versions

August 21st, 2026

This addendum to the Release Notes for code versions `3.0.10` and `3.1.2` (structured/narrow grids), and `3.2.1` (unstructured/wide grids) stems from the transition of the ITER SDCC cluster default environment from the `2023b` toolchain to `2025b`, as well as the move of most of the code documentation to MS Teams and GitHub. It also contains all other code updates mature enough to be passed on to the *master* version since the original 3.0.10 release on GitHub. The AD code was also updated to match the 3.2.1 standard code version. Changes listed here apply to all three code versions, unless specifically noted otherwise. The code online documentation and installation instructions have been updated to reflect the changes detailed below. The CI test cases and associated scripts have also been modified according to these code changes.

Run environment and compilation improvements:

- The code now can be compiled in full multi-threaded mode, across its various components. This can be achieved, interactively with the `-jN` option to *gmake*, or, if using the batch compilation scripts (such as *itermake* and *efgwmake*), with a `-N`<*number_of_processes*> option. The compilation preparatory steps (`listobj`, `depend`, `VERSION`, `local`, etc...) have been grouped in a new `prereqs(_debug)` target at the top *Makefile* level. The submodule *Makefile*s were thus extensively modified to make sure that no race conditions will occur (although some may still exist, please report them if you encounter any). If provided multiple targets, the batch compilation scripts will compile each target sequentially (with each target being internally multi-threaded) and will stop if a target fails to compile, before attempting the next one.
- The code now supports compilation with the IntelLLVM *ifx* compiler. However, because of a known bug in its OpenMP implementation that is fatal for **B2.5** runs, when attempting to compile the code with *ifx* and OpenMP options, the *Makefile* will attempt first to revert the compiler to *ifort*, if still available.
- When updating the code, the update scripts detect whether a valid **ADAS** repository address is available for download and if a pre-existing download is complete before attempting to sync with the remote repository. If not, the data files are fetched from the Open-ADAS website and a check file added if the downloads completed successfully and in full.
- The minimal `CMake` version for compiling **Eirene** has been increased to `3.10`.
- Some additional support for the Linux ARM system on individual laptops is also provided.
- The *setup.csh* file now accepts a *setup.csh.$HOST_NAME.$COMPILER.pre* for instructions to be given before it starts the bulk of its work. This can be used for instance to enforce centralized installation settings. Other user-transparent improvements related to centralized installations are also included.
- A new `SOLPS_DISABLE_ENV_CACHE` environment variable is available to turn off the reading and writing of the local environment files normally produced by *setup.csh*.
- The small utility program written by the *Makefile* to self-detect which MPI library is available has been improved to no longer cause compilation errors that could confuse users when looking at the compilation logs.
- The detection of the Motif library by the *setup.csh* script to decide whether to compile **DivGeo** has been improved to avoid some false positives, and the **DivGeo** *Makefile* makes a better use of that detection result.
- A new `PARAVIEW_MAJOR_VERSION` preprocessor variable has been introduced, to allow for proper linking to the Catalyst debug engine in ParaView, which uses different Fortran-to-C wrappers in version 6 compared to previous versions. The ParaView version number is automatically detected by the *Makefile* and used to compile the *cxxAdaptor.cxx* file within **B2.5**. If ParaView is not present on the system, the variable takes a default value of `0`.
- The default set of loaded modules for the Intel and `gfortran` toolchains on the `ITER` SDCC now corresponds to the `2025b` toolchain, meaning using the *ifx/2025b* API and/or the GCC/14.3.0 compiler. This also means updates of the `IMAS-Fortran` Access Layer to version `5.6.0`, IMAS Data Dictionary to `4.1.1`, `IMAS-ParaView` to `2.4.0`, `IDStools` to `2.4.1`, `Doxygen` to `1.14.0`, the `MSCL` library to `1.2.5`, `CMake` to `4.0.3`, `SimDB` to `0.15.2`, and `JSON`-Fortran to `9.0.5`. A new environment variable, `SOLPS_TOOLCHAIN_VERSION` can be used, if set before the `source setup.csh` command, to choose the `2023b` toolchain.
- On the `ITER` SDCC, if the NAG library module is unavailable, the Intel MKL library is searched for instead, before resorting to OpenBLAS.
- On the `ITER` SDCC and `ITM` EUROfusion Gateway (EFGW) machines, the `-warn interfaces` flag is added to the **B2.5** Intel debug compilations by default. If an interface error occurs, because of the changed interface to a routine `B`, called by a routine `A` that is compiled first in alphabetical order, an additional dependency can be explicitly added in the *$SOLPSTOP/modules/B2.5/config/dependencies.local* file, with the line:
  `${OBJDIR}/A.o: ${OBJDIR}/B.o`
- The `ORNL` configuration files have been extensively overhauled to reflect the new HPC architecture available, and to take full advantage of their centralized installation.
- A `g2` account setting has been added to the `ITM` SLURM job submission scripts.
- The two *csh* and *ksh*-based OpenMP setup scripts have been made consistent with each other.
- New compiler options have been implemented that improve the **Eirene** performance (when using an Intel compiler), and, if compiled with *ifort* (NOT *ifx*), remove the penalty that used to exist when running with a small number of OpenMP threads, and provide the expected level of scaling.
- The `SOLPS_MPIRUN` environment variable can be set to overwrite the default MPI launcher instructions used by *b2run* when submitting a run.
- Lots of safeties were sprinkled through in many configuration files and scripts against undeclared environment variables and/or missing executables.
- A new set of configuration files and a job submission script for the `SCNET` machine in China are available (for the Wide Grids code only). The `DLUT` and `LEUVEN` configuration files and `ORNL` hostname detection rules have also been updated.
- The code has been modified in several areas to remove some outstanding compiler warnings, with no loss in functionality.

Updates to code Input/Output:

- The `IMASDIR` environment variable can now be used more flexibly to redirect the IMAS code output from the code or specify the location of input IDSs.
- New target quantities are written to the *summary* IDS when creating an IMAS data entry: the peak electron temperature, the peak electron density, and the density-weighted average electron temperature along each target. Also the `gas_injection_rates` node now supports the DT and B species. In the Wide Grids code, a bug was corrected that affected the output of the power fluxes to the divertor targets.

- A bug was fixed to handle the writing of *fort.44* fields when the number of surfaces in either block 3A (non-standard surfaces) or 3B (additional surfaces, i.e. walls) of the **Eirene** input file is zero. Block 3B being empty can happen in the Wide Grids code workflow, if only the vacuum vessel contour was defined as part of the walls for **Eirene** in the **DivGeo** model.
- Another bug was fixed to prevent a spurious zeroing of the `LKINDP` array, which provides the correspondence between the **B2.5** species list and the plasma background species list given to **Eirene** in block 14 of the **Eirene** input file.
- The **b2co** program can now deal with comment lines in the *b2mn.dat* input file.
- The **b2us** program (only present in the Wide Grids version) has been corrected so the *b2mn.dat* file is rewound before the transport coefficient settings are read, since the code switches need to be accessed first.
- When using the tangent or Hessian AD code, the residuals of the parameter dependencies are rescaled by their `PAR_RESCALE` parameter (Wide Grids version only).
- The output of the atom and molecule profiles along the second target to the *b2time.nc* file have been corrected and the output logic generalized for cases with only one divertor target. The output of the neutral diffusivity for the AFN model has also been fixed (applies to the Wide Grids code version only).
- The *reshe* and *reshi* scripts have been modified to show the correct residuals when dealing with multi-species runs.
- The extrapolation rules for plotting **Eirene** reaction rates within the **b2plot** and **amds** programs have been corrected.
- Some bugs in **b2plot** related to the detection of the first closed flux surface have been corrected (applies to the Wide Grids code version only).
- The *b2fplasma* output for the `rcxna` and `rcxmo` fields has been corrected (applies to the Wide Grids code version only).
- Several bugs in the conversion program **b2yt** have been fixed, that affected the mapping of limiter cases, handling of BGK species, converting to a smaller grid, and reading reaction cards from **Eirene** MsV-formatted input files.
- The **nkn2dg** program for the equilibrium converter from the "new" Naka format to **DivGeo** was revived.
- The Doxygen build of the *equtrn/* manual inside **DivGeo** was updated to version `1.14.0`, but with a backward-compatibility option to still support older versions.
- Various minor corrections and clarifications of output and error messages are sprinkled in.

Corrections to the physics model:

- When using `BCPOT=11`, it is now possible to apply an iterative scheme to the potential equation to ensure better convergence, using the `b2news_max_iter_po` switch (applies to the Wide Grids code version only).
- The optimization code (Wide Grids version) can now use the BFGS method. The Hessian code can be invoked by using the `-hess_tgt` option to *b2run*.
- Errors were corrected for boundary conditions `BCENE/I=18`, `BCENI=24`, `BCENI=25`, and `BCENZ=16`, in the Wide Grids code, where the boundary condition computed and/or applied was from the first cell of the boundary, instead of all of them.
- Bug fixes were applied to boundary conditions `BCENI=24` and `27` (which only affected cases where the boundary was split into several regions).
- An index error was corrected that affected the `rcxmo` tally calculation, and another for the electron heat source due to volume recombination. The latter only affected the Wide Grids code version.
- The `prl_cur` switch, multiplier to the parallel current, is now only applied to the parallel current calculation proper, not also to the parallel conductivities `alf` and `sig` (affects the Wide Grids code version only).

Notes for developers:

- The **Uinp** common blocks have been modularized. The *ubpcom.h*, *uib2pprm.inc*, and *uinpcom.inc* and *uinput.inc* files have been ported into *ubpcom.F*, *uib2pprm.F*, and *uinput_mod.F*, respectively.
- Just as per the structured code, the *b2stbc_phys.F* routine in the Wide Grids code has been split internally with contained subroutines for each equation type. The local sound speed calculation has been moved to a stand-alone contained function to ensure consistency of definition across boundary condition types.
- The AD optimization code was made more compatible with multiple versions of the PetSC/TAO library.
- The arrays in *b2mod_input_profile.F* are now allocated dynamically before reading the prescribed transport coefficient profiles in *b2.transport.inputfile*, to allow for arbitrary radial resolution (Wide Grids code version only).
- The **EDGE2D** user routines for **Eirene** have been updated.